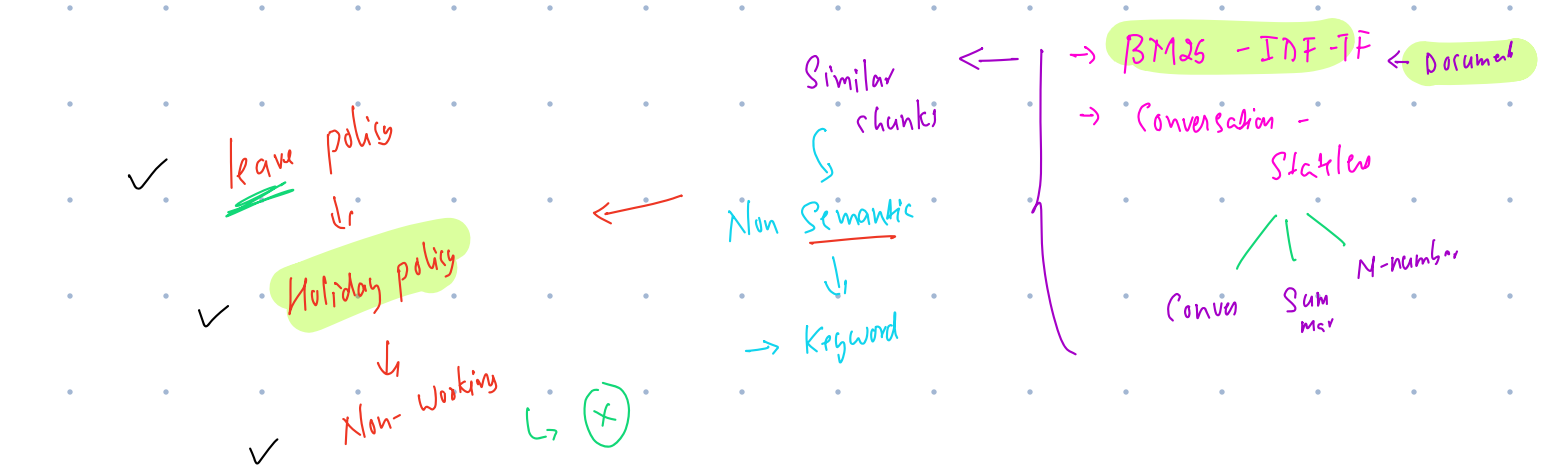
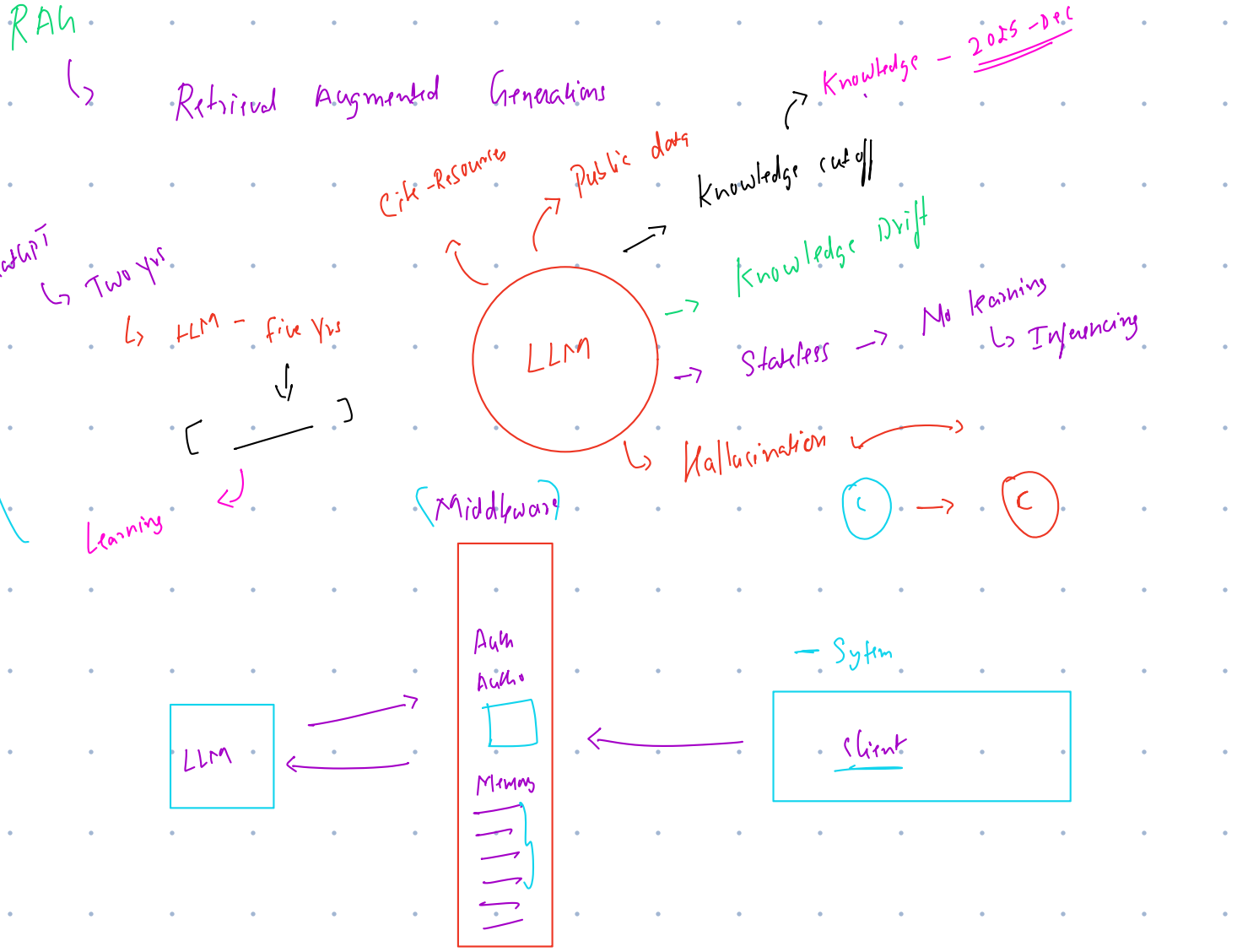
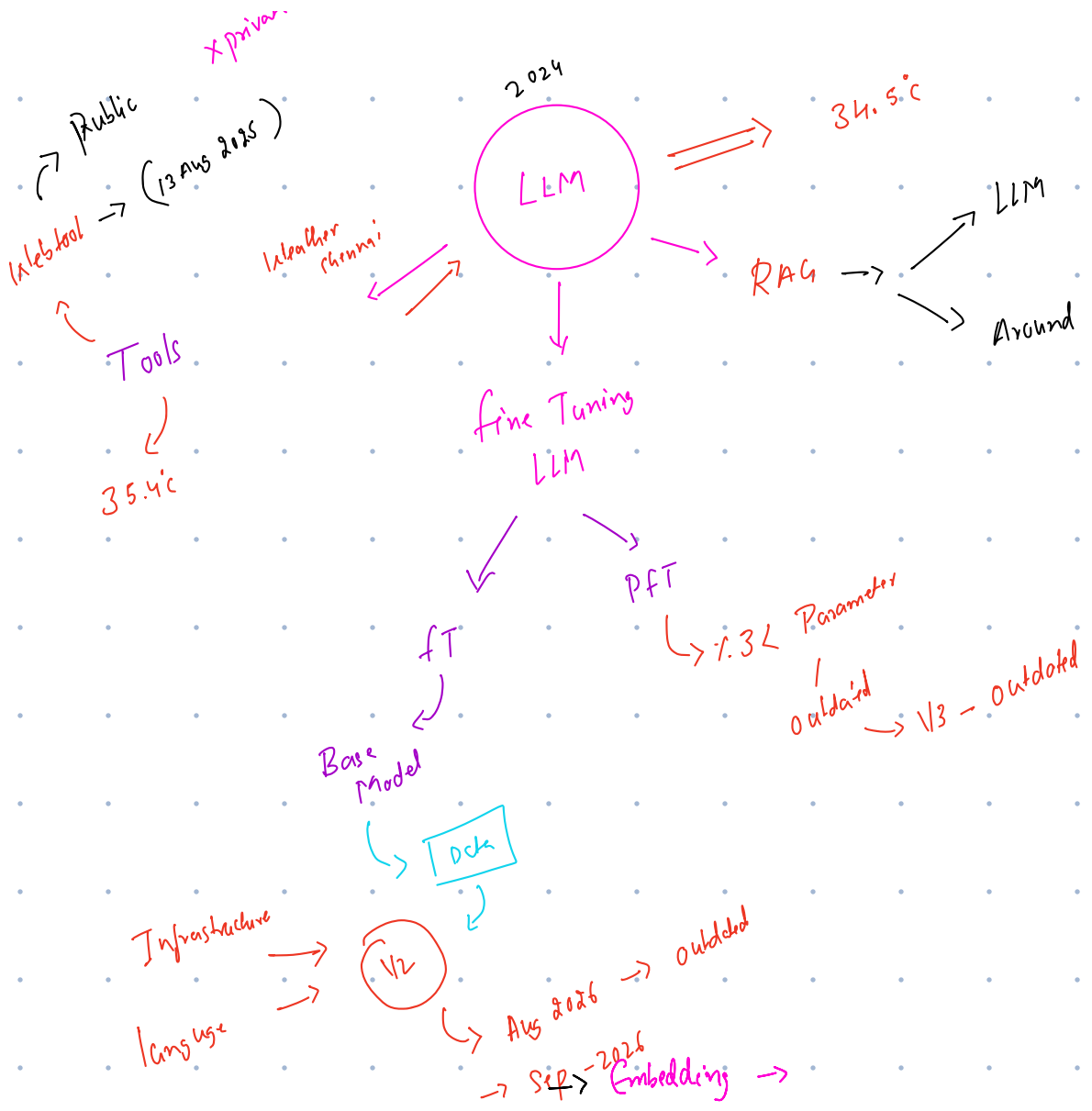



Org -> Push -> Accountability -> Reliability



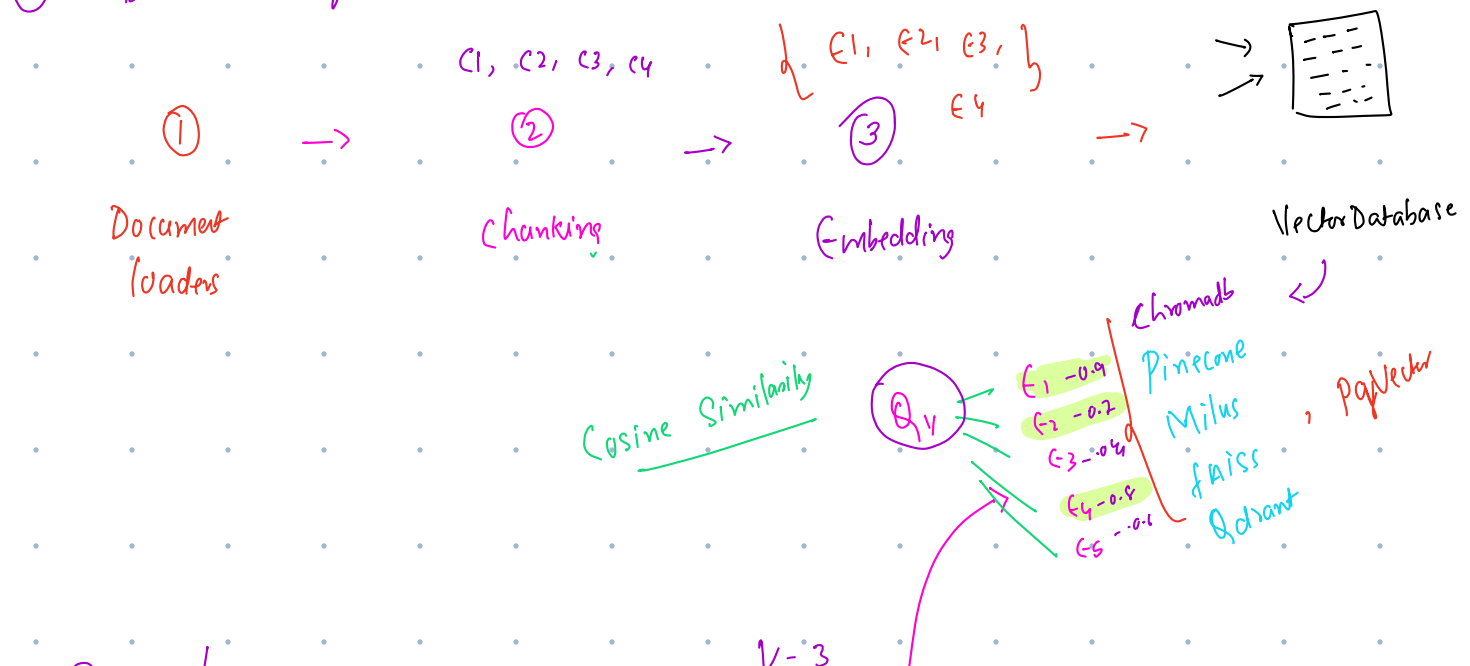

R etrival
↳
Seek / find
↳ knowled
↳ KB

A
↳
Augmet

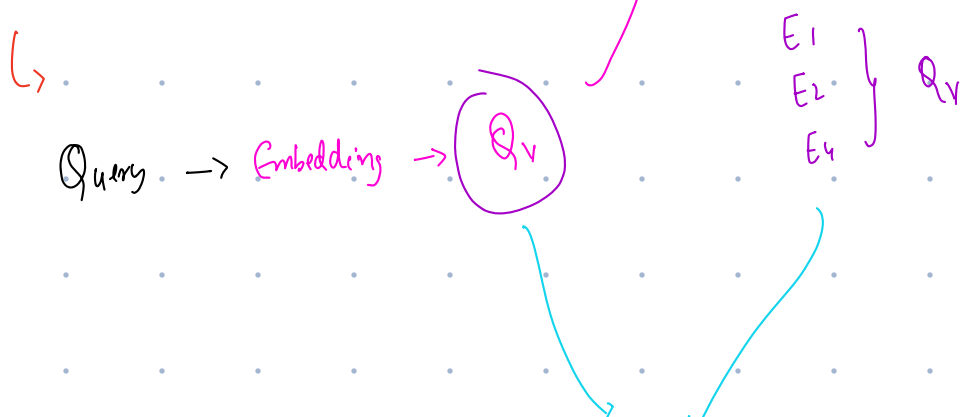
G
↳ LLM - find
Answer

Query → LLM → Answer

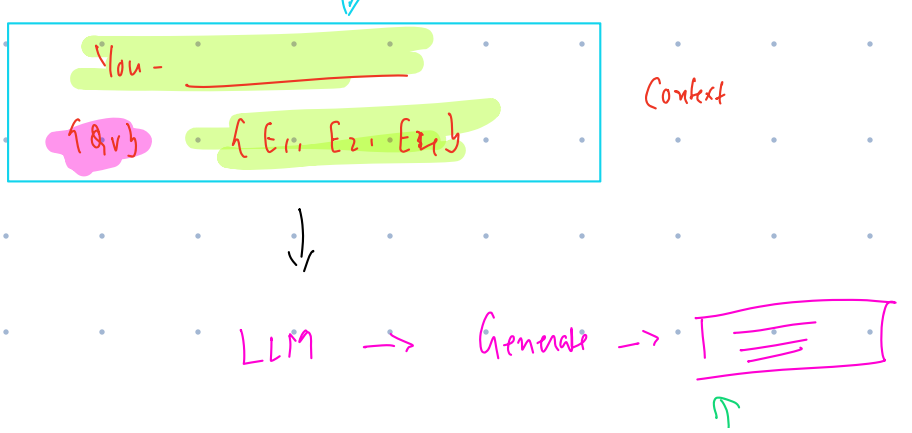

## ① Data Ingestion

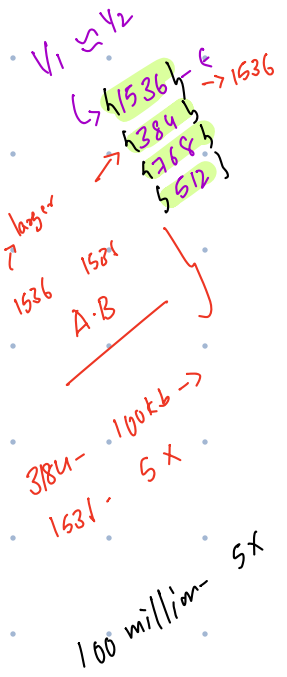



## ② Querying/ Retrieval



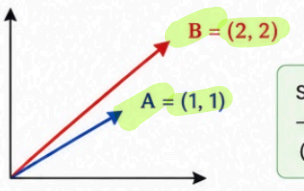

## ③ Generation





## 1. THE IDEA

Two vectors can have different lengths but point in the same direction.
We should consider them 100% similar.




Same direction → similarity = 1 (100% similar)

## 2. COSINE SIMILARITY FORMULA

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \, \|\mathbf{B}\|}$$

- $\mathbf{A} \cdot \mathbf{B}$ = dot product (captures angle and length)
- $\|\mathbf{A}\|, \|\mathbf{B}\|$ = magnitudes (lengths) of the vectors

Dividing by the magnitudes removes the effect of length and leaves only the angle (direction).

## 3. EXAMPLE: WHY MAGNITUDE MATTERS

**Using Dot Product (without magnitude)**

Let $\mathbf{A} = (1, 1)$ and $\mathbf{B} = (2, 2)$

Dot product:

$$\mathbf{A} \cdot \mathbf{B} = 1 \times 2 + 1 \times 2 = 4$$

The value 4 is large, but it doesn't tell us whether the vectors are in the same direction or just large in size.

**Using Cosine Similarity (with magnitude)**

Magnitudes:

$$\|\mathbf{A}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\|\mathbf{B}\| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

Cosine Similarity:

$$\frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \, \|\mathbf{B}\|} = \frac{4}{(\sqrt{2})(2\sqrt{2})} = \frac{4}{4} = 1$$

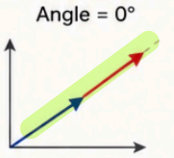
Cosine similarity = 1 → They are perfectly aligned (same direction)!

## 4. INTUITION: THINK OF ARROWS

**Magnitude** = how long the arrow is.

**Direction** = where the arrow points.

**Same Direction**
Angle = 0°

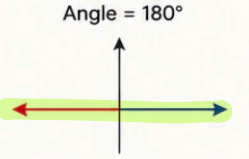


Cosine Similarity = 1 (Perfectly similar)

**Perpendicular**
Angle = 90°

Cosine Similarity = 0 (No similarity)

**Opposite Direction**
Angle = 180°

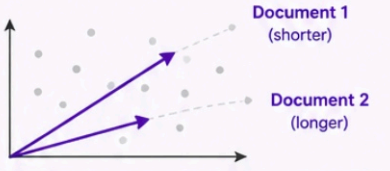


Cosine Similarity = -1 (Completely dissimilar)

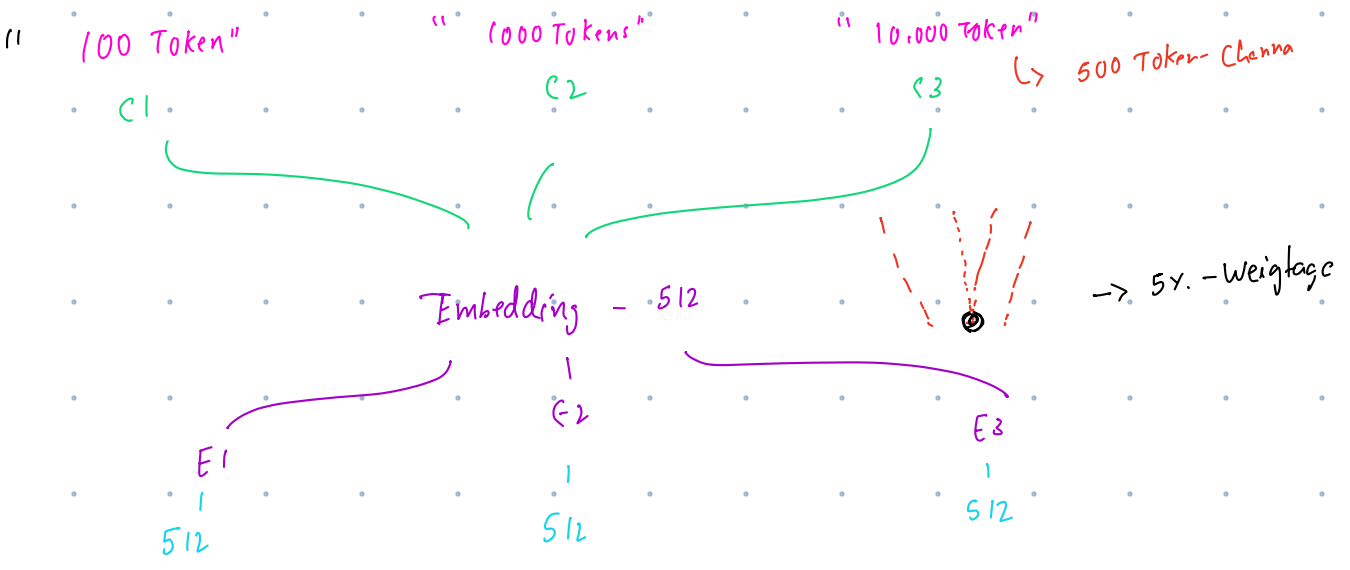
## 5. WHY IS THIS USEFUL?

In real-world data (like text embeddings):

- Two documents can have different lengths or overall scales, but still be about the same topic.
- Cosine similarity compares their directions, so it identifies **semantic similarity, not size.**




Similar direction → High cosine similarity



## How Much - Embedding Model

| Model | Dimensions | Max Tokens | Best For | Cost |
|---|---|---|---|---|
| text-embedding-3-small (OpenAI) | 1536 | 8191 | General use, production | Low (API) |
| text-embedding-3-large (OpenAI) | 3072 | 8191 | High-accuracy tasks | Medium (API) |
| text-embedding-ada-002 (OpenAI) | 1536 | 8191 | Legacy, widely used | Low (API) |
| all-MiniLM-L6-v2 (Sentence-Transformers) | 384 | 256 | Fast, local, low cost | Free (local) |
| all-mpnet-base-v2 (Sentence-Transformers) | 768 | 512 | High quality local | Free (local) |
| bge-large-en-v1.5 (BAAI) | 1024 | 512 | Best open-source quality | Free (local) |
| e5-large-v2 (Microsoft) | 1024 | 512 | Strong retrieval perf. | Free (local) |
| Cohere embed-english-v3.0 | 1024 | 512 | Reranking-friendly | Low (API) |
| Voyage AI voyage-2 | 1024 | 4096 | Long context | Low (API) |

MiniLM6 V2

| Use Case | Recommended Chunk Size | Overlap | Strategy |
|---|---|---|---|
| General Q&A over documents | 400-600 tokens | 50-100 tokens | Recursive Character |
| Code retrieval | Full functions/classes | 0 (function boundary) | Language-aware splitter |
| Legal / contract analysis | 300-500 tokens | 100 tokens | Semantic or Structure-aware |
| Medical literature | 200-400 tokens | 50 tokens | Semantic Chunking |
| Customer support FAQ | 100-200 tokens | 20-50 tokens | Sentence-level |
| Long research papers | 500-800 tokens | 100-150 tokens | Semantic or Header-based |
| Real-time fact retrieval | 50-150 tokens (propositions) | 0 | Proposition-based |

MySQL → B-Tree

Vector Database

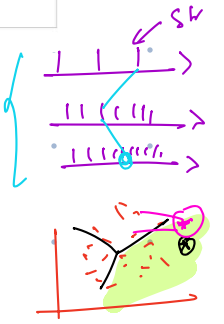
→ Retrieval

→ Indexing

→ features

| Database | Deployment | Scale | Hybrid Search | Metadata Filter | Best Use Case |
|---|---|---|---|---|---|
| | process) | | | | |
| Chroma | In-process or server | < 1M | No | Yes | Dev, small projects |
| Pinecone | Cloud managed | 1B+ | Yes (sparse) | Yes | Managed production |
| Weaviate | Self-hosted / cloud | 100M+ | Yes (built-in BM25) | Yes | Hybrid search prod |
| Qdrant | Self-hosted / cloud | 100M+ | Yes (sparse vectors) | Yes (rich) | High-perf production |
| pgvector | PostgreSQL extension | 10M | With extensions | Full SQL | Postgres shops |
| Milvus / Zilliz | Distributed / cloud | Billions | Yes | Yes | Enterprise scale |

| Index Type | Recall | Speed | Memory | Build Time | Best For |
|---|---|---|---|---|---|
| Flat (Exact) | 100% | Slow (O(N)) | Low | Instant | < 100K vectors |
| HNSW | 95-99% | Very fast | High | Moderate | Most production RAG |
| IVF-Flat | 90-98% | Fast | Low-Medium | Moderate | 100K-10M vectors |
| IVF-PQ | 85-95% | Very fast | Very low | Slow | 10M+ vectors, memory constrained |
| ScaNN (Google) | 98% | Fastest at scale | Medium | Slow | Extreme scale |



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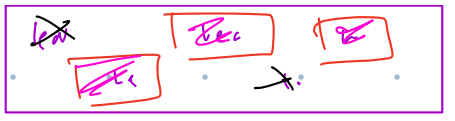
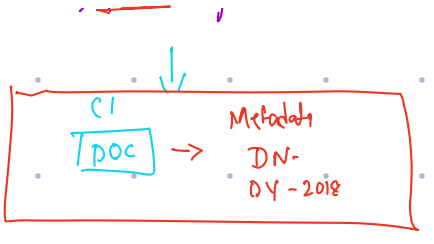
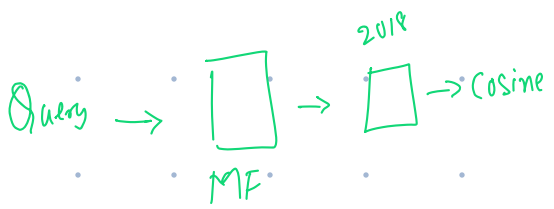


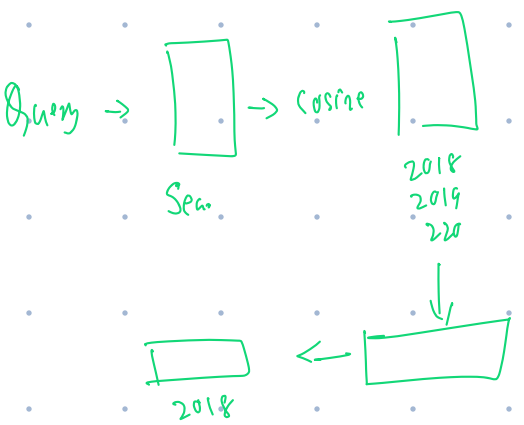


Two places

Before Retrieval




After Retrieval




Query - Embeddings → Text 3 small

↳ 1536

$\frac{A \cdot B}{|A|\,|B|}$

1536 .384

Vectors - VDBs

Min 1642 → 384

↳ 10000 |-

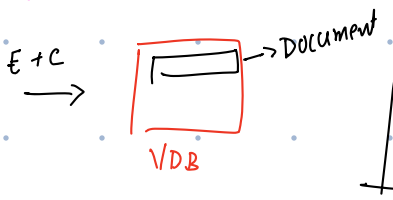
↳ 384

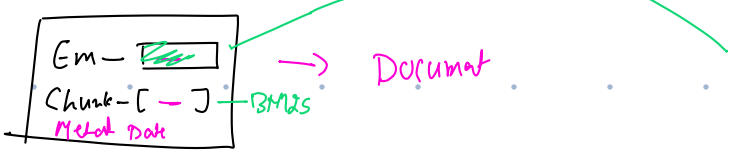
→ LLM

C1, C2, C3 →

↳ EM

e1 c2 c3




Ve

Embedding